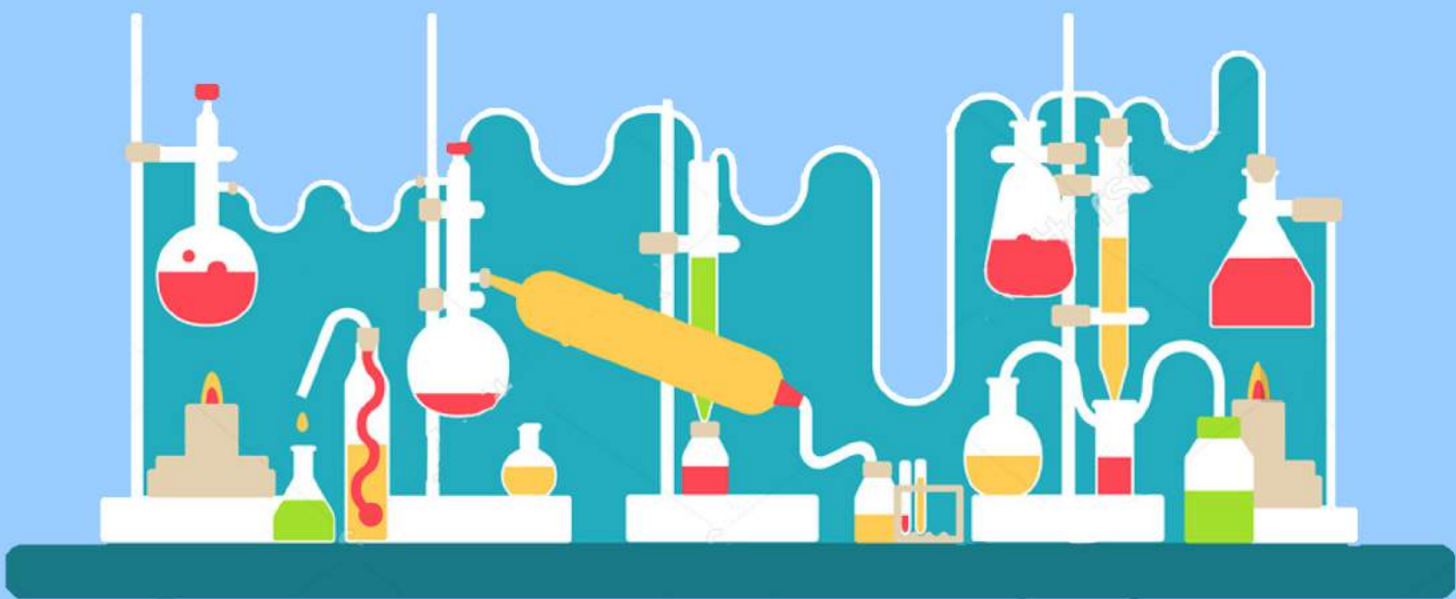
BIOLOGY LABORATORY

لجنة الطب البشري
فكرة تحيا على نبض قلوبكم

لجنة طب الأسنان
Dental Committee
معا نرسم الابتسامة

#3 Marcomolecules

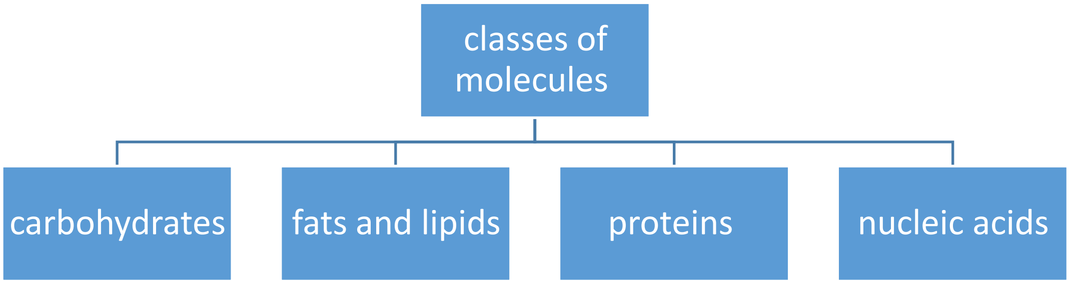


نضعُ بين أيديكم هذا العملَ الّذي كان نتاجًا للتّعاون بين (لجنة الطبّ البشريّ) و (لجنة طبّ الأسنان)، و نُوجِّهه لطلّاب السّنة الأولى (دُفعة 2016) كمصدرٍ رئيسيٍّ و شاملٍ لمادّة (مختبر البيولوجي) ، و نَضمنُ و فيه شرحٌ لكم اشتمالَه على المطلوبِ منكم من الكتاب و السلايدات، موضِّح للأفكارِ المهمّة، و قد بذلنا جهدنا أن نُخرجه لكم بالرّجوع للمصادر الموثوقة، هذا و قد تجدون بعض التوضيحات باللّغة العربيّة 3: تيسيرًا عليكم؛ إذ هذا أدعى لترسيخ الفهم

تعاون زملاؤكم على إخراجه بصورته المُتقَنة إذ أدركوا مسعى الإتقانِ و مقصدَه، فكان أن وَضعوا أمام أعينِهم صورةَ الطّبيب الّذي ستكونونه كلُّكم، فلم يرضَوا إلّا بأن يكونَ طبيبًا حقًّا، تقديرًا منّا للعلمِ و أهلِه، و إيمانًا بأنّ بذرةَ الخيرِ لا تُنبِتُ إلّا خيرًا :")

كلّ التوفيق ^^"

All organisms consist of chemicals. These molecules are made of atoms, some of these are molecules small in size such as water molecules, but others are large such as proteins and polysaccharides.



These biological macromolecules are important cellular components and perform a wide array of functions necessary for the survival and growth of living organisms.

Macromolecules may exist as <u>polymers</u> and <u>non-polymers(Monomers)</u>.

Polymers are large compounds made from small monomers bonded to each other by condensation reaction that a water molecule is released(dehydration), These polymers can be degraded to small monomers by hydrolysis reaction (water molecule is added to the bond).

Those reactions need <u>specific-enzymes</u> and in humans there are over 1000 enzymes are used to initiate deferent metabolic reactions. Either hydrolysis or dehydration.

✓ **Biological Molecules in human body:**

1- Carbohydrate:

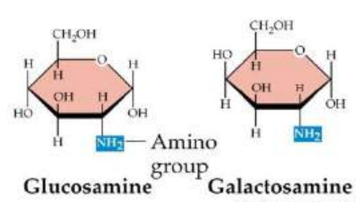
- are composed of carbon, oxygen and hydrogen.

- They have the same amount of carbon and oxygen and twice as much as hydrogen (O:H:C = 1:2:1) .

- Amino sugars: have nitrogen present as in “chitin”.

الصورة فقط للتوضيح .. ليست للحفظ

(b) Amino sugars

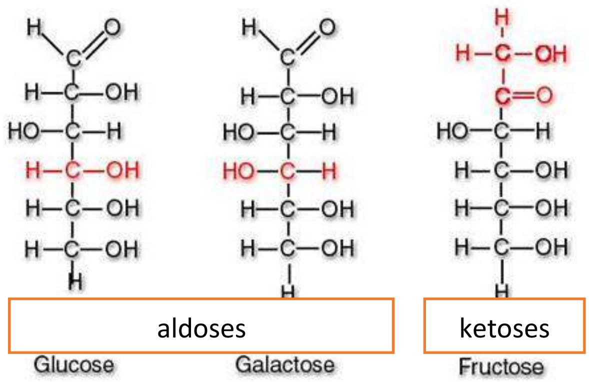


There are 2 types of of Carbohydrates:

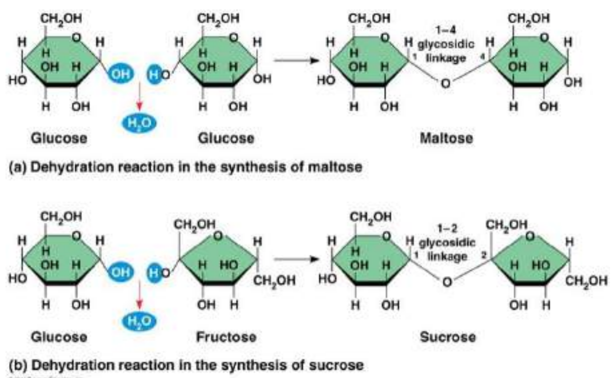
1-Simple: Monosaccharides & disaccharides.

2- Complex: polysaccharides.

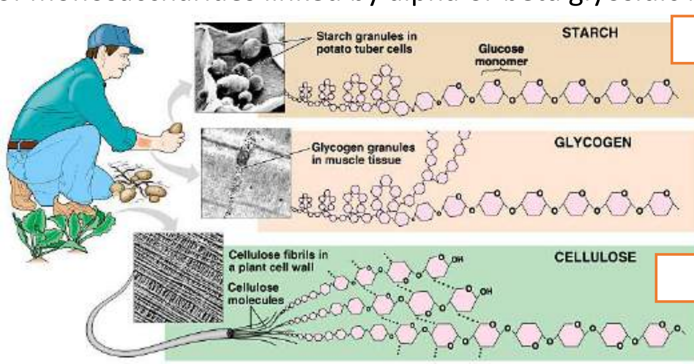
Monosaccharides: have aldehyde or ketone, Known as aldoses or ketoses



Disaccharides: 2 monosaccharides linked by glyosidic bond via condensation reaction (dehydration)



Polysaccharides: polymers of monosaccharides linked by alpha or beta glyosidic bond.



alpha glycosidic bond in starch

Beta glycosidic bond in cellulose

Identification of carbohydr

Benedict's Test is used :for reducing Sugars is a test which determines the presence of **reducing sugars** in a test solution.

The principal reagent in Benedict's Test for Reducing Sugars is Benedict's Solution which contains

- copper(II) sulphate
- sodium carbonate
- sodium citrate

-هذا المحلول يستخدم للكشف عن السكريات القابلة للأكسدة

- تذكر ان هذه السكريات عندما تتفاعل مع هذا المحلول فإنها تقوم باختزاله اما هي عن نفسها فيتم اكسدتها(تتأكسد).

What are reducing sugars?

-Sugars are classified as **reducing sugar** or **non-reducing sugar** based on their ability to act as a reducing agent during the Benedict's Test or not. A reducing agent donates electrons during a redox reaction (reducing-oxidizing reaction) therefore it becomes oxidized

Is any sugar that is capable of acting as reducing agentbecause it has a free aldehyde group or a free ketonegroup.[1] All monosaccharides are reducing sugars, along with somedisaccharides, oligosaccharides, and polysaccharides.

The monosaccharides can be divided into two groups: the aldoses, which have an aldehyde group, and the ketoses, which have a ketone group. Ketoses must first tautomerize(they must be in **in an open chain form**)to aldoses before they can act as reducing sugars.

The **aldehyde functional group** is the reducing agent in reducing sugars. Reducing sugars have either an aldehyde functional group or have a ketone group **- in an open chain form** - which can be converted into an aldehyde.

- Some examples of monosaccharides are glucose, fructose, ribose, xylose, glyceraldehyde, and galactose. Examples of reducing disaccharides are lactose and maltose. (these disaccharides have only one anomeric carbon involved in the glycosidic bond therefor they are capable to be converted to the open form)

-Note that the disaccharide **sucrose and trehalose is not a reducing sugar**. In fact, sucrose is the most common non-reducing sugar. These disaccharides stuck in cyclic form and cant be converts to the open chain form because of the (glycosidic bond between their anomeric carbons)

Is the Benedict's Test for reducing sugars qualitative or quantitative?

The test may be used as qualitative or quantitative.

-Qualitative : because its used to discover the existence of reducing sugars.

-Quantitative :because the color obtained correlates to the concentration of reducing sugars in the solution and that gives an idea about the quantity of sugar present in the solution, hence the test is semi-quantitative. A greenish precipitate indicates about 0.5 g% concentration; yellow precipitate indicates 1 g% concentration; orange indicates 1.5 g% and red indicates 2 g% or higher concentration.

يتضمن هذا التفاعل تحول ذرات ال Cu^{+2}(copper II) من محلول sodium citrate-------الى------- Cu^{+1}(copper I) على شكل مركب يترسب (اكسيد النحاس Cu_2O copperI oxide) و كلما ازدادت كمية هذا الراسب زاد تغير اللون الازرق (لا تفاعل) الى اللون الاحمر القرميدي (تفاعل عالي)

تذكر من التجربة

1 - ان ال sucrose لا يتفاعل و تكون نتيجته سلبية (لون ازرق) لانه =

stuck in the cyclic form therefor can't change to the open chain form.

2 - ال starch لا يمكن ان يتفاعل (لون ازرق) لانه ليس من reduced sugars و عديد السكريات polymer

3 - blue <greenish <yellow< orange<red

What is theLugol's Iodine Test for Starch?

-Remember that: starch is an polysaccharide with beta glycosidic bond

The Iodine Test for Starch is used to determine the presence of starch in biological solutions or samples.

-The test is qualitative only.

Lugol's Iodine Test consists of iodine dissolved in an aqueous of potassium iodide

amylose or the straight chain of starch	deep blue- black color
Amylopectin or the branched portion of strach	Orange/ yellow hue color

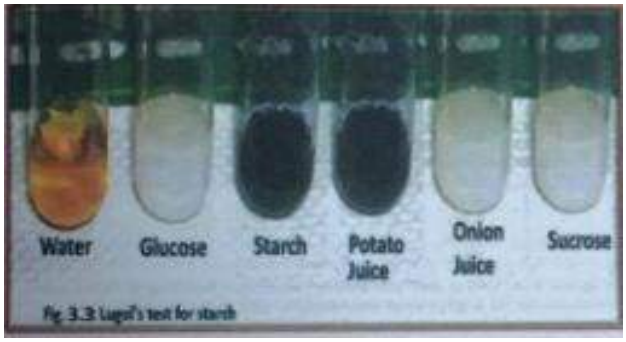


Fig. 3.3 Lugol's test for starch

Fats & lipids:

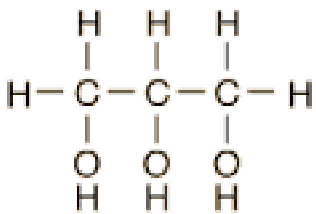
- Lipids are made of the elements **Carbon, Hydrogen** and **Oxygen**,
- (Hydrophobic Molecules) although they have a much lower proportion of water than other molecules such as Carbohydrates. They are insoluble in water.

- Lipids perform many functions, such as:
  - Energy Storage
  - Making Biological Membranes
  - Temperature insulation
  - Protection - *e.g. protecting plant leaves from drying up*
  - Buoyancy
  - Acting as hormones

1-Fats:

They are made from two molecules: **Glycerol** and **Fatty Acids**.

A Glycerol molecule is made up from <u>three Carbon</u> atoms with a **Hydroxyl Group** attached to it and Hydrogen atoms occupying the remaining positions.

Glycerol	**Fatty Acids**
A Glycerol molecule is made up from three Carbon atoms with a **Hydroxyl Group** attached to it and Hydrogen atoms occupying the remaining positions.  Glycerol	Fatty acids consist of an (Carboxyl group “Acid Group” –COOH) at one end of the molecule and a Hydrocarbon Chain(hydrophobic chain) , which is usually denoted by the letter '**R**'. Fatty acids may be **saturated** or **unsaturated**.

- Fatty acids consist of an **Acid Group** at one end of the molecule and a Hydrocarbon Chain, which is usually denoted by the letter '**R**'.

- Fatty acids may be **saturated** or **unsaturated**. A fatty acid is saturated if every possible bond is made with a Hydrogen atom, such that there exist no C=C bonds. Unsaturated fatty acids on the other hand do contain C=C bonds. Obviously monounsaturated fatty acids have one C=C bond, and polyunsaturated have more than one C=C bond.

- If fatty acids are unsaturated, their shape is altered from a saturated molecule so the molecules in the Lipid push apart, thus making it more fluid and oily.

- Animals tend to have more saturated, and consequently solid at room temperature lipids whereas plants tend to have more unsaturated and so fluid at room temperature lipids.

Triglycerides:

- Triglycerides are lipids consisting of one glycerol molecule bonded with three fatty acid molecules. The bonds between the molecules are covalent and are called **Ester bonds**. They are formed during a **condensation reaction (dehydration reaction)**.

- Triglycerides are **hydrophobic** and so insoluble in water. The charges are evenly distributed around the molecule so hydrogen bonds to not form with water molecules.

Phospholipids:

- Phospholipids are similar to triglycerides in they consist of a glycerol 'backbone' and fatty acid 'tails', however, the third fatty acid has been replaced by a phosphate group 'head'.

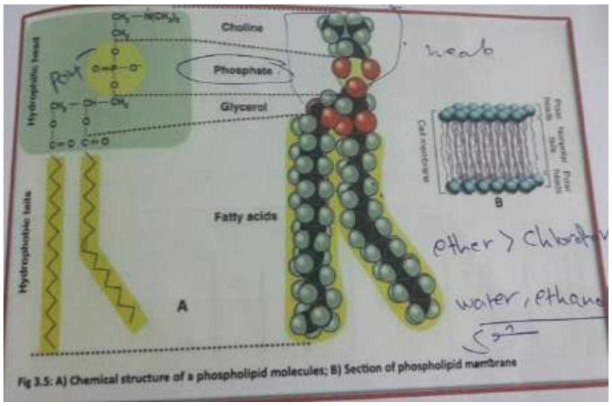


Fig 3.5: A) Chemical structure of a phospholipid molecules; B) Section of phospholipid membrane

- While the fatty acid 'tails' are hydrophobic, the phosphate 'head' is hydrophilic. This means the phosphate group will orientate itself towards water and away from the rest of the molecule .
  and also gives rise to the special properties that allow phospholipids to be used to form membranes.

- Phospholipids can contain saturated and unsaturated fatty acids. This allows for the control of the fluidity of membranes, which is useful, for example, in maintaining membrane fluidity at low temperatures.

Steroids:

- steroid core structure is composed of seventeen carbon atoms, bonded in four "fused" rings. steroids vary by the functional groups attached to this four-ring core and by the oxidation state of the rings. steroids have two principal biological functions: certain steroids (such as cholesterol) are important components of cell membrane which alter membrane fluidity, and many steroids are signaling molecules which activate steroid hormone receptors.

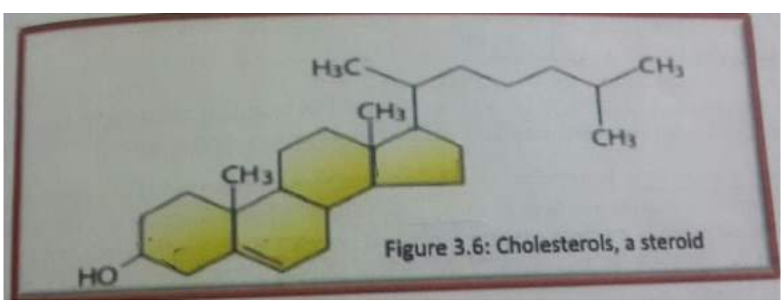
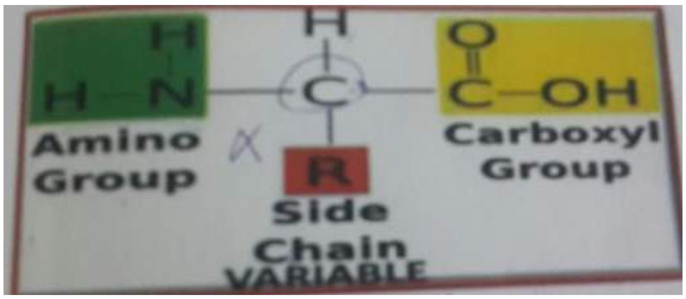


Figure 3.6: Cholesterols, a steroid

- The solubility of fats and lipid in different solvent depends on the polarity of the solvent, when the polarity of the solvent increases the solubility increase. **The most polar solvent is ether then chloroform**, while water and ethanol show the lowest polarity.

Proteins:



They act as structural components and enzymes, they are a polymer of amino acid, we have a 20 different amino acid. And all have amine group & carboxyl group are bonded covalently to the central carbon atom which is called alpha carbon. Also there are H atom and R group are bonded to carbon.

Amino acids joined together by dehydration with releasing of water, and the covalent linkage here is peptide.

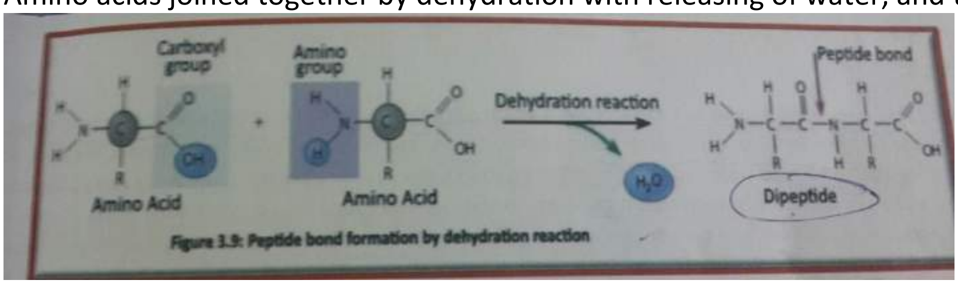


Figure 3.9: Peptide bond formation by dehydration reaction

What is the Biuret Test for Proteins?

The **Biuret Test** is often used to determine the presence of **peptide bonds** in protein. As a student you will most likely will be testing for the presence of protein in foods.

The Biuret test for proteins may also be **extended** to quantitatively measure the **concentration of total protein** using spectrometric methods.

The Biuret reagent is the sole reagent in the Biuret Test for proteins. The Biuret reagent contains

- Hydrated Copper sulphate
- Potassium hydroxide solution
- Potassium sodium tartrate

Violet color indicates the presence of peptide bonds

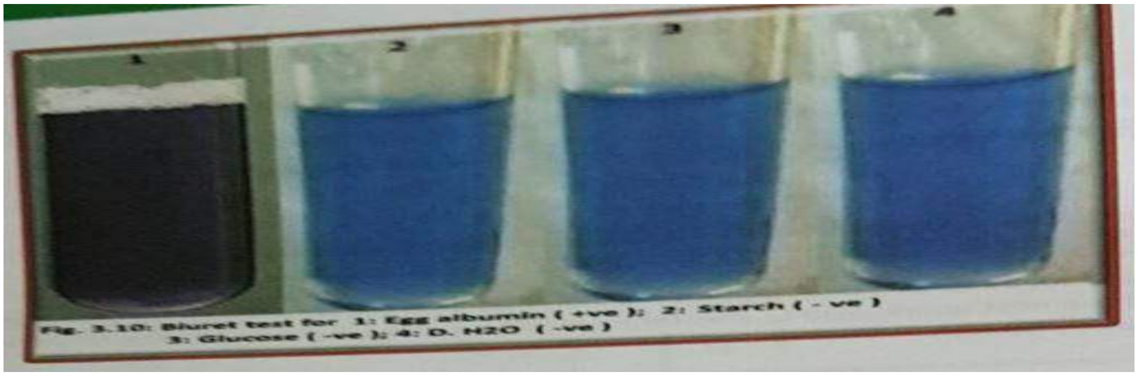


Fig. 3.10: Biuret test for 1: Egg albumin (+ve); 2: Starch (- ve) 3: Glucose (-ve); 4: D. H2O (-ve)

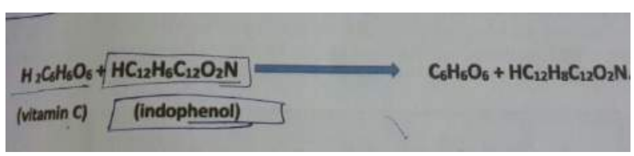
Vitamin C:

Ascorbic acid, water soluble vitamin, it forms connective tissue, bone, teeth, blood vessel wall. Diet deficient in vitamin C may lead to scurvy. Also it is anti-oxidant

Any food contains little or no Vitamin C will turn brown after exposure air like apple, and the food that contain Vitamin C will not so that it is anti-oxidant.

Humans don't make their own Vitamin C like some plants and animals. So they need sources to get it. The Vitamin presents in food such citrus food, green vegetables. Vitamin C has chemical instability because of it is a strong reducing agent. And it can be deactivated by wide range oxidizing agents like atmospheric oxygen.

The experiment studies the concentration of Vitamin C in lemon juice and grapefruit and other juices. Vitamin c is reducing agent and weak acid so it s concentration in the juices will be determined by the oxidation and reduction titration:



In this reaction, the reactants and products are colorless or pale yellow with the exception of indophenol, which is either blue (pH 4), or purple (pH S 4), Hence, the reaction between vitamin C and indophenol bleaches indophenol, and the color of the reaction mixture can be used to detect the end point of the titration then determine the vitamin C concentration in the juice.

The end.

Test your self ☺

Q1/: *According to the experiments of biomolecules of living things complete the following table:*

Organic molecules	Reagent/s	Color of (+ve) result	Color of (-ve) result
Starch	iodine (lugol)	Blue black	red
Protein	NaOH+CuSO4	violet	blue
(reducing) ~~su~~ Aldehid sugers	Benedict's	red, ~~orange~~	Blue
Vitamin C	Indophenol	Colorless	blue

Q2: *Circle the best correct answer:*

1- **In Biuret test reveals the presence of:**
A- H-bond B- Covalent bond C- Peptide bond D- Ionic

2- **Starch is a polymer of :**
A- Galactose B- Glycerol C- Glycogen D- Glucose

3- **Indophenol is the indicator to quantities:**
A- Ascorbic acid B- Amino acid C- Starch D- Glucose

4- **Benedict have specific characteristics :**
A- Qualitative B- Quantitative C – is CuSO4 D- All of Above

Report Solutions:

Carbohydrates

Table 1: Benedict's Test for Reducing Sugars

Tube	Solution	Initial Color	Color of the mixture	Remarks
1	5% Glucose	blue	orange to red	high amount of reducing sugar
2	5% Sucrose	blue	blue	it doesn't contain reducing sugar
3	5% Starch	blue	blue	it doesn't contain reducing sugar
4	Onion juice	blue	yellow	it contains reducing sugar
5	Potato juice	blue	green	it doesn't contain reducing sugar
6	Distilled water	blue	blue	it doesn't contain reducing sugar

Aldehyde

Table 2: Lugol's Test for Starch

Tube	Solution	Initial Color	Color of the mixture	Remarks
1	5% Glucose	brown	Brown	it doesn't contain starch
2	5% Sucrose	brown	brown	it doesn't contain starch
3	5% Starch	brown	Dark blue	it contains starch → poly
4	Onion juice	brown	brown	it doesn't contain starch
5	Potato juice	brown	Dark blue	it contains starch
6	Distilled water	brown	brown	it doesn't contain starch

Lipids and Fats

Table 3: Solubility Test for fat

Tube	Solution	Solubility of oil	Remarks
1	Ether	Soluble	it is non-polar same as the oil
2	Acetone	Soluble	it is non-polar same as the oil
3	Ethanol	insoluble	it is polar while oil is not
4	Chloroform	Soluble	it is non polar same as the oil
5	Distilled water	insoluble	it is polar while oil is not so it formed two layers

Proteins

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Table 4: Biuret Test for Peptide bond

Tube	Solution	Initial Color	Color of the mixture	Remarks
1	Egg albumin solution	Light blue	violet	it contain peptide bond (proteins).
2	2.5% Glucose	light blue	light blue	it doesn't contain peptide bond.
3	1% Starch	light blue	light blue	it doesn't contain peptide bond.
4	Distilled water	light blue	light blue	it doesn't contain peptide bond.

Vitamin C

Table 5: qualitative test of vitamin C in different juices:

Juices	No. of juice need to make the indophenol colorless	Rank of Juice according to [vitamin C]	
1- Lemon	4	The highest	it has a high concentration of [vitamin C] so it needed the least amount of juice
2- Orange	8	The medium	it has a medium concentration of [vitamin C] so it needed a normal amount of juice
3- Grapefruit	10	The lowest	it has a low concentration of [vitaminC] so it needed the highest amount of juice